AgriConnect

A Freshness Aware Agricultural Marketplace

Software Requirements Specification

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1. INTRODUCTION

1.1 **Purpose**

This Software Requirements Specification(SRS) document outlines and defines functional and non-functional requirements for AgriConnect, an agricultural marketplace designed to reduce post-harvest losses among farmers in Ghana by connecting the farmers directly to buyers without the need for middlemen.

1.2 **Intended Audience**

- Course Supervisor
- Software Development Team
- Project sponsors

1.3 **Scope**

AgriConnect is an agricultural marketplace that connects Ghanaian smallholder farmers directly to verified buyers and truck drivers.AgriConnect empowers farmers by minimizing post-harvest losses and facilitating a direct connection with buyers, thereby preventing the financial exploitation of middlemen. The system specification includes an on-device AI crop freshness detection model, a farmer produce listing, buyer marketplace, automated transport dispatch, payment processing and a tamper-proof transaction audit trail.

2. OVERALL DESCRIPTION

2.1 **Product Perspective**

AgriConnect is a mobile application that introduces a software coordination layer into Ghana's existing but fragmented agricultural supply chain. The system does not replace any physical infrastructure. It organizes and connects farmers with produce, buyers with demand and truck drivers.

Current Process:

- Farmers harvest produce, wait for middlemen to arrive and accept whatever price is offered under time pressure.
- No objective quality signal exists to support price negotiation or remote buyer trust.
- Buyers have no direct visibility into available produce, quantity or freshness across regions.
- Transport is arranged through personal contacts and informal networks with no coordination mechanism.
- Payments are made in cash with no verifiable record of the transaction chain.

Proposed System:

- Farmers can scan produce at harvest using an on-device AI model that returns a verified freshness score and estimated shelf life.
- Verified listings are published to a buyer marketplace with AI-generated price recommendations based on live regional market data.
- Buyers browse and purchase directly via Mobile Money escrow, with funds released only on confirmed delivery.
- The nearest available truck driver is automatically dispatched via SMS upon purchase confirmation.
- Every transaction is recorded in a tamper-proof audit trail.

2.2 **Product Features**

- AI Crop Freshness Detection
- Produce Listing and Price Recommendation
- Buyer Marketplace
- Payment Processing
- Automated Transport Dispatch
- Tamper-Proof Audit Trail

2.3 User Classes and Characteristics

Farmer

Responsibilities:

- Scan produce using the AI freshness detection model at point of harvest.
- Create product listings on the marketplace.
- Confirm driver pickup and receive Mobile Money payment on delivery.

Skill Level:

- Limited formal education assumed for a significant proportion of users.
- Basic smartphone familiarity required.
- No prior experience with digital marketplaces assumed.
- Platform designed for minimal text interaction with visual cues as primary navigation aid.

Buyer

Responsibilities:

- Browse and filter produce listings on the marketplace.
- Purchase produce via different payment options.
- Confirm delivery to trigger farmer payment release.

Skill Level:

- Basic smartphone and Mobile Money literacy assumed.
- Commercially oriented users market traders, restaurant operators, food processors and institutional buyers.
- Expected to engage with the platform regularly and independently.

Truck Driver

Responsibilities:

- Receive automated dispatch notifications.
- Accept or decline delivery jobs.
- Confirm delivery completion to trigger payment release.

Skill Level:

- Familiarity with Ghanaian road networks across the operating region assumed.
- Registered in advance with vehicle capacity and operating region on file.

System Administrator

Responsibilities:

- Verify and approve farmer, buyer, and driver account registrations.
- Maintain overall platform integrity and user account management.

Skill Level:

- Basic computer literacy with web browser access

2.4 Operating Environment
User Side

- Android/iPhone smartphones running Android 8.0/iOS 16.0 or higher.
- Minimum 2GB RAM on client device.
- Device camera required for AI freshness scan functionality.

Server Side

- Linux-based cloud server hosted on Render.com
- Node.js runtime environment
- PostgreSQL database hosted on Supabase

AI Model

- Runs entirely on-device via TensorFlow Lite
- No server-side inference required
- Compatible with ARM-based Android processors

Network

- Internet connectivity required for marketplace, payment and SMS dispatch features
- AI freshness scan operates fully offline

2.5 Assumptions and Dependencies

Assumptions

- Farmers have an Android smartphone with a functional camera.
- All registered users have access to a Mobile Money account on MTN or Vodafone Cash.
- Users have access to basic internet connectivity for all features excluding the AI freshness scan

Dependencies

- Paystack API for Mobile Money payment processing and escrow management.
- Arkesel Ghana SMS API for automated driver dispatch notifications.
- TensorFlow Lite runtime for on-device AI model inference.
- Supabase for PostgreSQL database hosting.
- Render.com for backend API hosting and deployment.

3. SYSTEM FEATURES

3.1 Functional Requirements

Functional Requirement	Description
User Login	Users should be able to log in with their username and password.
User Sign up	Users should be able to sign up as a farmer, driver or buyer.
User Browsing	Users should be able to browse through the product listings.
User Payment	Users should be able to pay for a listing through a payment platform.
Farmer Payment	Farmers should be able to receive payment from the app through a payment platform
AI Scan	The system shall allow farmers to scan harvested produce using the device camera and generate a freshness score and estimated shelf life using AI.
System Notification	The system shall notify farmers when the freshness score is projected to fall below a predefined threshold within 48 hours.
User Profile	The system shall allow users to edit and update their profile information.
Purchase Notification	Farmers should be able to see when their produce has been bought.
Produce Upload	Farmers should be able to upload multiple images of their produce.

3.2 Non-Functional Requirements

Non-Functional Requirement	Description
Performance	1.The AI freshness scan shall return results within 5 seconds. 2. The system shall load the home page within 3 seconds under normal network conditions. 3.Marketplace search results shall be displayed within 2 seconds for standard queries.
Privacy	The system shall comply with applicable Ghanaian data privacy and financial transaction regulations.
Usability	The system shall provide an intuitive user interface that allows first-time users to complete product listing or purchase with minimal guidance.
Reliability	The application shall support at least 1000 concurrent users without significant performance.
Security	The application shall encrypt all sensitive user information stored in the database.